City University of Hong Kong

Master of Science in Data Science

SkinSmith: A Constraint-Aware Generative Agent for Game Weapon Skin Design - A Counter-Strike 2 Case Study

Registered Project Topic: Latent Vision → Image Generation

SDSC6002 Research Project

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Abstract

Text-to-image models produce rectangular images, whereas a usable game-asset texture must remain coherent after it is cut into a fragmented UV atlas, mapped onto a narrow three-dimensional surface, viewed from several directions, and exported under asset-specific constraints. This report presents SkinSmith, a human-supervised generative Agent that converts a short theme keyword into a mapped and reproducible weapon texture. The system introduces three checkpoints: the user confirms an expanded visual theme, selects one textual art direction, and selects one artwork only after seeing its source image together with left, right, and top weapon previews.

The technical method separates a tileable-pattern baseline from a weapon-aware route. The latter derives a canonical coordinate frame from the target mesh, constructs per-UV-pixel position and normal maps, fits a selected master artwork to continuous weapon-space canvases, and then bakes the result into the asset's existing UV atlas. Candidates are evaluated using texture statistics, a mesh-derived seam metric, and fixed multi-view renders. Selection first enforces hard constraints and then applies Pareto dominance and a deterministic objective order. One local refinement round is permitted, but it is retained only when it improves the locked score by at least 0.01; otherwise the system rolls back.

In a four-pair formal experiment on an AK-47 asset, the weapon-space route reduced asset-seam error by a mean of 0.2693 and improved the multi-view score by 0.0601, winning all four paired comparisons on both measures. A separate selection-policy ablation showed that an equal weighted score chose an infeasible seam width, whereas constraint-first selection returned a valid operating point. All four formal refinement attempts were rejected and rolled back, demonstrating bounded no-regression behavior rather than guaranteed self-improvement. A real three-checkpoint run generated four mapped artwork alternatives from the keyword *dragon*; the user-selected candidate was exported as a 2048 by 2048 RGB texture with asset-seam error 0.00078 and multi-view score 0.8166. Transfer to an M4A4 adapter reused the core pipeline without new image generation, but did not satisfy the AK-47 retention threshold, exposing the asset-specific limits of portability.

The results support the value of composing and evaluating artwork in the target asset's geometry rather than treating image generation as the final output. They also show why human selection, explicit feasibility rules, preserved evidence, and rollback are necessary in a creative Agent whose automatic metrics remain imperfect.

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1 Introduction

1.1 Problem context

Generative image models have made it straightforward to produce visually rich concept art from a short text prompt. That capability does not, by itself, solve the problem of designing a texture for an existing three-dimensional game asset. The visible surface of a weapon is stored as a set of UV charts: pieces of the surface are flattened, rotated, scaled, and packed into a square image. A source illustration that looks coherent in two dimensions can therefore be split across unrelated atlas regions, disappear on narrow components, or become inconsistent between the left, right, and top views. Export requirements, paintable regions, texture resolution, and seam behavior add further constraints.

This distinction matters for both research and practice. If the generated source image is treated as the result, an evaluation can reward an attractive rectangle while overlooking the failures introduced by mapping. Conversely, a simple tileable pattern may survive arbitrary UV fragmentation but provide little control over whole-weapon composition. A useful system must connect creative generation to the actual asset contract and must make its decisions observable.

SkinSmith addresses this integration problem. It is a constraint-aware generative Agent for weapon-skin design, evaluated through a Counter-Strike 2 case study. The system accepts a registered weapon and a compact theme such as *dragon*. It expands the theme into a visual world, offers several textual directions, generates several landscape artworks for the selected direction, and maps every artwork to the target geometry before asking the user to choose. The chosen source is then processed at full resolution, evaluated, optionally refined once, and exported with prompts, model identifiers, hashes, metrics, and decisions.

1.2 Scope

The research object is the Agent workflow and its geometry-aware texture pipeline, not a universal Counter-Strike skin generator. The AK-47 is the complete case study. An M4A4 adapter is used to examine transfer, but it is not presented as an equivalent production result. The system preserves each asset's existing UV layout; it does not perform automatic UV unwrapping. Its software renderer provides controlled technical views rather than photorealistic game-engine rendering. Steam Workshop publication is outside the project scope.

The group originally registered under the umbrella topic *Latent Vision → Image Generation*. Its initial brief proposed a broad investigation of image-to-image generation, diffusion models, prompt engineering, conditioning, and creative outputs such as illustrations, game concept art, and posters. With fewer than three weeks remaining, the supervisor advised the group to narrow the work to the technical development of a completed image-processing Agent and explicitly stated that it did not need to remain the original broad Latent Vision project. SkinSmith is the resulting scoped research project: it retains image generation and human-AI collaboration from the registered topic, but concentrates them on one technically testable asset pipeline.

This approved narrowing made it possible to evaluate the complete path from intent to deployable texture, including negative results, UV failures, selection errors, and rollback behavior that would be invisible in a source-image-only study. The registered topic is therefore retained on the title page for administrative continuity, while the report title names the actual system and research question.

1.3 Research questions

The study addresses five questions:

RQ1 Does composing artwork in a mesh-derived continuous weapon space and then baking it to the authoritative UV atlas improve mapped asset-seam and multi-view performance relative to a generic tileable-pattern baseline?

RQ2 What failures become observable when candidates are evaluated after mapping to the real weapon from left, right, and top views rather than only as source images or flat texture atlases?

RQ3 Can hard-constraint filtering, Pareto-based selection, and a bounded correction with rollback prevent an Agent from accepting technically invalid or lower-quality updates that a soft aggregate score might prefer?

RQ4 Can three explicit human checkpoints separate theme intent, art-direction choice, and mapped-artwork selection while allowing the Agent to begin from one short keyword?

RQ5 Which parts of the pipeline transfer to a second weapon through a `GameAssetAdapter`, and which geometry, material, camera, and deployment contracts remain asset-specific?

1.4 Contributions

The project makes five bounded contributions:

1. A weapon-space-to-UV generation route that composes selected artwork in a canonical frame derived from the target mesh and bakes it through per-UV-pixel geometry maps into the existing atlas.
2. Render-in-the-loop evaluation in which source images, true asset seams, and fixed left, right, and top renders are retained as separate evidence.
3. A small-pool multi-objective policy that separates feasibility from preference and permits one local correction with an explicit rollback gate.
4. A three-checkpoint human-Agent interaction that preserves human authority over the theme, direction, and mapped artwork.
5. An adapter boundary that distinguishes reusable planning and evaluation logic from asset-specific mesh, UV, camera, and export contracts.

These claims concern the integration and evaluation of a deployable workflow. They do not claim the first use of diffusion for mesh texturing, the first multi-view texture method, or general compatibility with every game asset.

1.5 Report structure

Section 2 positions the work against diffusion, texture synthesis, 3D-aware texturing, constrained selection, and human-AI systems. Section 3 describes the Agent, asset representation, weapon-space bake, evaluation, and refinement rules. Section 4 defines the controlled experiments and evidence protocol. Section 5 reports findings in the order of the research questions. Section 6 considers validity, limitations, project management, and responsible use before section 7 concludes.

2 Related Work

2.1 Image generation and structural control

Diffusion models learn to reverse a gradual noising process and provide a stable foundation for high-quality image synthesis (Ho et al., 2020). Improved parameterizations and sampling methods made the approach more efficient (Nichol and Dhariwal, 2021), while latent diffusion reduced the cost of high-resolution synthesis by operating in a learned latent space (Rombach et al., 2022). Joint image-text representations such as CLIP enabled general semantic comparison between prompts and images (Radford et al., 2021). These developments explain why an existing image model can serve as a strong source-art generator in SkinSmith.

The limitation is spatial control. A text prompt can describe a dragon, cloud, or metallic surface, but it does not specify how those elements should survive a particular UV atlas. ControlNet demonstrates the broader value of supplementing text with spatial conditions such as edges, depth, and segmentation (Zhang et al., 2023). SkinSmith adopts the same motivation without training a new conditioning network: semantic planning is separated from deterministic geometry-based placement and evaluation.

2.2 Texture synthesis, periodicity, and UV atlases

Classical texture-synthesis methods grow or assemble images from local neighborhoods and patches (Efros and Leung, 1999; Efros and Freeman, 2001). Graph-cut textures improve joins between overlapping patches by optimizing the cut through the overlap (Kwatra et al., 2003). Periodic-plus-smooth decomposition addresses discontinuities caused when a finite image is treated as periodic in the Fourier domain (Moisan, 2011). This work supports the use of a tileable pattern as a meaningful baseline and motivates the frequency-domain border repair used by Route A.

Square-border periodicity is nevertheless different from continuity on a mesh. Surface parameterization flattens a three-dimensional surface into one or more charts and must trade distortion against chart layout (Floater and Hormann, 2005). Least-squares conformal maps are a widely used approach to low-angle-distortion atlases, but chart boundaries still interrupt large regular patterns (Lévy et al., 2002). TextureMontage further shows that positioning multiple images on an arbitrary surface requires explicit control over distortion and transitions (Zhou et al., 2005). In a game pipeline, the UV contract is often already fixed. SkinSmith therefore does not replace the parameterization; it derives geometry maps and seam correspondences from the bound asset and works within that atlas.

2.3 Diffusion-based 3D texturing

Several systems use two-dimensional diffusion priors to generate appearance for three-dimensional objects. DreamFusion demonstrated that rendered views can connect a text-to-image prior to a 3D representation without a large paired 3D dataset (Poole et al., 2022). TEXTure progressively paints an existing mesh from depth-conditioned views and supports text-guided generation and editing (Richardson et al., 2023). Text2Tex similarly uses a progressive generate-and-refine process with view partitioning and automatic view selection (Chen et al., 2023). TexFusion aggregates denoising predictions in a shared latent texture representation to improve global consistency (Cao et al., 2023). Point-UV Diffusion combines a low-frequency point-space representation with UV-space refinement (Yu et al., 2023). Paint3D adds UV inpainting and high-resolution refinement to produce complete, lighting-less texture maps (Zeng et al., 2024). Synchronized multi-view diffusion shares denoised content between overlapping views during sampling (Liu et al., 2024).

This literature establishes that UV discontinuity, view inconsistency, incomplete coverage, and baked illumination are central problems. It also limits the novelty claim that can reasonably be made here: SkinSmith is not the first system to texture a mesh using diffusion. Its focus is a different layer of the problem—an asset-bound, human-supervised Agent that begins with compact intent, retains the target game's UV contract, exposes mapped alternatives before formal execution, enforces deployment constraints, and preserves the evidence behind every decision.

2.4 Rendering and evaluation

Differentiable rendering research formalizes the link between a 3D representation and image-space supervision (Kato et al., 2018; Liu et al., 2019). SkinSmith does not optimize through rendering gradients, but follows the same general principle: the mapped views are evidence about the texture that cannot be recovered from the source image alone.

Perceptual metrics also require care. LPIPS showed that learned feature distances often correlate with human perceptual judgments better than simple pixel distances (Zhang et al., 2018). CLIPScore provides reference-free semantic compatibility, while documenting cases in which such a score is weaker (Hessel et al., 2021). Accordingly, SkinSmith keeps texture, seam, multi-view, and semantic evidence separate. Automated review recommends and explains; it does not replace artwork selection by the user.

2.5 Constrained selection and bounded Agents

Weighted aggregation is convenient, but a high score on one objective can hide a violation on another. Feasibility-oriented constraint handling instead ranks valid solutions ahead of invalid ones (Deb, 2000). NSGA-II formalizes non-dominated sorting for multi-objective optimization

Table 1: Positioning against the closest 3D-texturing systems. "Not central" means that the property is not a principal reported contribution, not that an extension would be impossible.

Work	Main mechanism	Consistency and UV treatment	Human and deployment control
TEXTure	Iterative depth-conditioned painting	Successive views are projected to a mesh texture	Text-guided generation and editing; hard deployment gates are not central
Text2Tex	Progressive generation and refinement	View partitioning and automatic next-view selection update visible texels	Prompt-controlled; no explicit rollback contract reported
TexFusion	Shared latent texture denoising	View predictions are aggregated on one latent texture	Text-conditioned; no staged human checkpoint contract
Point-UV	Point-space coarse colour followed by UV diffusion	Low-frequency global condition plus UV refinement	Conditional generation; no game-specific export gate
Paint3D	Multi-view fusion, UV inpainting, and UVHD refinement	Complete high-resolution lighting-less UV maps	Text or image conditioning; focuses on texture quality rather than Agent rollback
SyncMVD	Synchronized denoising across overlapping views	Shares latent content at each denoising step	Text-guided; no three-stage selection contract
SkinSmith	User-selected master artwork and deterministic weapon-space bake	Preserves the authoritative UV; evaluates fixed mapped views and true seams	Three checkpoints, hard feasibility, one bounded correction, and rollback

(Deb et al., 2002). SkinSmith adapts these principles to a small, expensive candidate pool rather than running an evolutionary search: it filters hard violations, computes the feasible Pareto set, and uses a fixed objective order.

Agent research provides a second relevant foundation. ReAct interleaves reasoning and actions so observations can change the trajectory (Yao et al., 2023). Self-Refine and Reflexion use feedback and memory to improve later attempts without updating model weights (Madaan et al., 2023; Shinn et al., 2023). These methods motivate explicit observation, diagnosis, action, and memory, but they do not imply that every revision is better. SkinSmith therefore limits refinement to one local round and requires measured improvement before acceptance.

2.6 Human control and responsible generation

Human-AI interaction guidelines emphasize communicating capabilities, allowing correction, and preserving user control when model behavior is uncertain (Amershi et al., 2019). Co-creative systems treat generation as a turn-based collaboration (Guzdial and Riedl, 2019), while PromptChainer demonstrates the value of visible intermediate stages for complex AI tasks (Wu et al., 2022). These ideas inform the three checkpoints in SkinSmith: theme, direction, and mapped artwork.

Finally, datasheets and model cards show why provenance, intended use, evaluation conditions, and limitations should be recorded (Gebru et al., 2021; Mitchell et al., 2019). Responsible-AI scholarship also warns that fluent outputs can obscure data, bias, environmental, and accountability concerns (Bender et al., 2021). The project consequently logs model and provider identifiers, redacts API keys, preserves formal runs, prohibits copied skins and protected characters, and

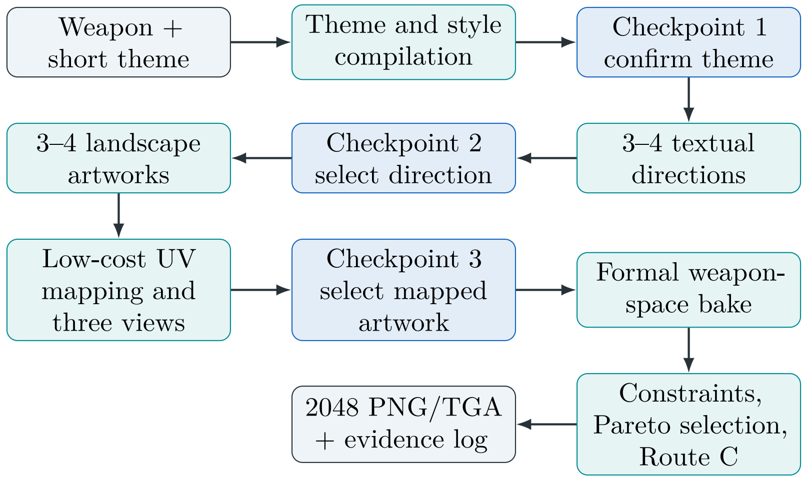


Figure 1: The SkinSmith workflow. The three blue nodes are blocking human checkpoints; formal processing cannot begin before a mapped artwork is selected.

keeps publication outside the tested scope.

2.7 Research gap

Prior work provides strong components: text-to-image generation, spatial conditioning, periodic texture synthesis, 3D-aware texture generation, multi-objective optimization, and iterative feedback. The gap examined here is their practical integration around an existing game asset:

> How can compact user intent be converted into a human-selected source design that is deterministically bound to an authoritative UV contract, evaluated after real geometry mapping, and refined only within explicit feasibility and rollback rules?

This framing treats deployability, evidence, and human authority as part of the method rather than as post-processing around a generated image.

3 Methodology

3.1 Design principles

The method was developed around four principles. First, generation is an intermediate operation rather than the final product. Second, every technical decision is made against a versioned asset contract. Third, feasibility and preference are separated during selection. Fourth, human approval is required at points where automatic metrics cannot adequately represent intent or aesthetic judgment.

Figure 1 summarizes the resulting pipeline. A thin client calls the same Agent runtime used by the command-line interface. The runtime coordinates planning, generation, asset processing, rendering, evaluation, and persistence; the user interface does not reimplement those operations.

3.2 Agent state and human checkpoints

An `AgentRunRequest` contains the asset identifier, a short theme or free brief, an optional broad style family, the candidate count, and explicit budgets for image calls, retries, and refinement rounds. The persisted state follows:

$$\texttt{created} \to \texttt{planning} \to \texttt{awaiting\_theme} \to \texttt{awaiting\_direction}$$
$$\to \texttt{awaiting\_artwork} \to \texttt{executing} \to \texttt{completed}.$$

At the first checkpoint, a `ThemeCompiler` expands a compact keyword into a validated `ThemePack`. The pack contains the concept, palette, primary and supporting elements, environmental and material cues, atmosphere, component story, evaluation criteria, and prohibited content. The user may confirm or edit this interpretation. A `StyleCompiler` then produces three or four materially different textual directions for the confirmed theme and selected weapon. At the second checkpoint, the user chooses one direction.

The selected direction generates three or four dense landscape master artworks. Each is validated for resolution, tonal and colour variation, local detail density, and forbidden content. It is then passed through a low-cost mapping stage using the target adapter. The original image and the left, right, and top views are treated as one indivisible candidate card. The third checkpoint asks the user to select one card. This design prevents a visually attractive source from being chosen without seeing how it behaves on the asset.

The selected source path and hash become part of a frozen design contract. Formal processing reuses that exact file; it does not regenerate an approximation after selection. Every transition, tool action, budget change, retry, and stop reason is written to persisted state so a failed or interrupted run can resume from the last valid checkpoint.

3.3 Asset adapter

Each target is represented by a versioned `GameAssetAdapter`. The adapter binds:

- the mesh path, version, and SHA-256 hash;
- the authoritative UV source and UV address mode;
- canonical longitudinal and up axes;
- semantic regions and creative anchors;
- paintable-area rules;
- fixed cameras;
- texture size and export format.

This boundary is important because mesh geometry, UV charts, and material behavior cannot be assumed to transfer between weapons. The accepted AK-47 adapter uses the new CS2 geometry, a 2048 by 2048 RGB texture, fixed left/right/top cameras, and a 24-bit TGA Custom Paint export. The M4A4 case supplies a different mesh hash, axes, UV topology, and seam calibration while reusing the core algorithms.

3.4 Mesh-derived geometry maps

Let a mesh triangle have three-dimensional vertices $\mathbf{p}_0, \mathbf{p}_1, \mathbf{p}_2$, vertex normals $\mathbf{n}_0, \mathbf{n}_1, \mathbf{n}_2$, and UV coordinates $\mathbf{u}_0, \mathbf{u}_1, \mathbf{u}_2$. For every covered atlas pixel $\mathbf{u}$, barycentric coordinates $(\lambda_0, \lambda_1, \lambda_2)$ are computed in UV space. The corresponding position and normal are

$$\mathbf{p}(\mathbf{u}) = \lambda_0 \mathbf{p}_0 + \lambda_1 \mathbf{p}_1 + \lambda_2 \mathbf{p}_2, \tag{1}$$

$$\tilde{\mathbf{n}}(\mathbf{u}) = \lambda_0 \mathbf{n}_0 + \lambda_1 \mathbf{n}_1 + \lambda_2 \mathbf{n}_2, \tag{2}$$

$$\mathbf{n}(\mathbf{u}) = \frac{\tilde{\mathbf{n}}(\mathbf{u})}{\|\tilde{\mathbf{n}}(\mathbf{u})\|_2}. \tag{3}$$

The mesh is transformed into a canonical frame whose x axis follows the weapon length, y is up, and z distinguishes the two sides. Positions are normalized to the asset bounds. Rasterizing all UV triangles produces a position map, a normal map, a valid-pixel mask, and a face-index map. These maps connect continuous weapon-space composition to the existing fragmented atlas.

3.5 Route A: tileable-pattern baseline

Route A generates a square, dense, crop-tolerant pattern. Its opposite image borders are repaired using periodic-plus-smooth decomposition (Moisan, 2011). The repaired square is then sampled through the original UVs and rendered on the weapon. This route is deliberately generic: it is expected to tolerate arbitrary crops, but it has no explicit whole-weapon hierarchy.

Two seam concepts are recorded. The square-border error measures differences between opposite image edges. The asset-seam error instead evaluates pairs of UV edges that correspond to the same welded three-dimensional edge. Passing the first measure does not imply passing the second.

3.6 Route B: weapon-space composition and UV bake

Route B uses one selected landscape master artwork. The source is fitted to continuous left, right, and top canvases in canonical weapon coordinates. The source must be dense and edge-to-edge, with several medium or small thematic clusters, because a single dominant subject or large empty region is unlikely to survive the narrow silhouette and UV fragmentation.

For each valid UV pixel, the canonical position determines the sampling coordinate on the side and top canvases. Surface normals blend the projections:

$$w_s = |n_z|^p, \qquad w_t = \max(n_y, 0)^p, \tag{4}$$

$$\mathbf{c}(\mathbf{u}) = \frac{w_s \mathbf{c}_s(\mathbf{p}) + w_t \mathbf{c}_t(\mathbf{p})}{w_s + w_t + \epsilon}, \tag{5}$$

Figure 2: The central Route B idea: compose in continuous weapon space, store fragments in the authoritative UV atlas, and judge coherence after reconstruction. The flat atlas is not expected to read as a conventional illustration.

where $p = 4$ in the accepted configuration. $\mathbf{c}_s$ is sampled from the appropriate left or right canvas according to surface orientation, while $\mathbf{c}_t$ is sampled from the top canvas. This soft blend avoids an abrupt switch between charts selected solely by the largest normal component.

3.7 True asset-seam metric and correction

OBJ files often duplicate vertices at UV boundaries. The method first welds coordinate-identical geometric vertices, identifies mesh edges shared by adjacent faces, and records pairs whose UV coordinates differ. These pairs form the true asset-seam graph.

For N paired samples with RGB colours $\mathbf{a}_i, \mathbf{b}_i$ and local gradients $\mathbf{g}_i^a, \mathbf{g}_i^b$, the colour and gradient terms are

$$E_c = \frac{1}{N}\sum_{i=1}^{N} \frac{\|\mathbf{a}_i - \mathbf{b}_i\|_1}{3 \cdot 255}, \tag{6}$$

$$E_g = \frac{1}{N}\sum_{i=1}^{N} \frac{\|\mathbf{g}_i^a - \mathbf{g}_i^b\|_1}{3 \cdot 255}. \tag{7}$$

The implementation combines these normalized terms into E_{asset} using the locked project weights. The principal hard requirement is $E_{\text{asset}} \leq 0.01$.

An edge-safety correction blends a narrow region around UV-island boundaries toward a stable base colour. Wider regions generally reduce seam error but can erase visible detail. Candidate widths are therefore evaluated against both the seam threshold and a minimum multi-view retention of 95% relative to the uncorrected texture.

3.8 Rendering and evaluation

The software renderer loads textured OBJ geometry, triangulates polygon faces, interpolates UV coordinates barycentrically, samples the texture bilinearly, and uses a z-buffer with fixed orthographic left, right, and top cameras. Lighting and cameras are deterministic so candidates are compared under the same conditions.

Texture-level evidence includes contrast, saturation, detail, paintable coverage, and square-border periodicity. The multi-view score combines foreground coverage, contrast, saturation, visible detail, and luminance consistency across the three views. The project also records mapped-element readability, defined as

$$S_{\text{read}} = 0.25 S_{\text{source}} + 0.50 S_{\text{left}} + 0.15 S_{\text{right}} + 0.10 S_{\text{top}}. \tag{8}$$

This weighting reflects the principal inspection view in the case study. It is recommendation-only and cannot override the user's artwork selection.

3.9 Constraint-first selection

For a candidate x, each hard constraint j has a non-negative violation $v_j(x)$. A candidate is feasible only if

$$V(x) = \sum_j v_j(x) = 0.$$

Feasible candidates are compared using Pareto dominance over the locked objectives. If several candidates remain non-dominated, a deterministic priority order selects the operating point. A conventional normalized weighted score is retained only as an ablation baseline. This policy prevents a strong visual score from compensating for a deployment failure.

3.10 Route C: bounded local correction

Route C begins from the selected Route B texture. Rendered component visibility masks identify a component whose detail level deviates from its target. The Agent creates exactly two correction candidates at fixed intensities. Changes are restricted to that component plus a transition halo, and the number of changed pixels outside the allowed region must be zero.

Each correction must retain all Route B hard constraints. Let S_B be the locked score of the original and S_C the best feasible correction. The correction is accepted only when

$$S_C - S_B \geq 0.01.$$

Otherwise the final result is the unchanged Route B texture. The method therefore tests diagnosis and action without assuming that automatic refinement is monotonic.

3.11 Generation backends and evidence

The generator interface supports a deterministic procedural baseline, a local SD-Turbo model, and provider-backed image generation. SD-Turbo is pinned to revision `b261bac6fd2cf515557d5d0707 481eafa0485ec2` at 512 pixels, two inference steps, and guidance scale zero. The accepted creative run used `gemini-3.1-flash-lite` for structured planning and `gemini-3.1-flash-image` for 1K landscape source images.

Every formal run preserves the request, prompts, provider and model identifiers, seeds where applicable, redacted traces, source hashes, validation, mapped views, UV outputs, metrics, decisions, budgets, retries, runtime, and checkpoints. API keys are read from environment variables and are not written to run evidence.

4 Experimental Design

4.1 Evidence protocol

Experiments were executed through repository scripts and preserved as immutable run directories. Report tables were extracted from machine-readable JSON and CSV records rather than copied from console output. Formal A/B/C evidence is kept separate from earlier technical smokes and from the edge-width sub-ablation. This separation is necessary because the candidate sources and experimental questions differ.

The principal asset is an AK-47 mesh with 26,133 triangles, 21,548 UV coordinates, 928 UV islands, and 4,999 welded-topology seam pairs. The accepted texture size is 2048 by 2048 RGB for final export. Formal comparison runs use 512-pixel working textures for controlled evaluation and share the target mesh, cameras, metrics, palette, candidate count, and base seeds.

4.2 Formal A/B/C comparison

The formal experiment contains four paired candidates with seeds 20260716 to 20260719:

Route A a generic pattern-first square source with periodic repair;

Route B a whole-weapon composition created in continuous weapon space, baked through OBJ-derived geometry maps, and corrected at UV edges;

Route C the corresponding Route B result plus one diagnosis and two local correction attempts, subject to the 0.01 improvement gate.

Route A and Route B intentionally use different route-specific generation logic. The comparison asks whether the weapon-aware representation produces better mapped behavior, not whether identical pixels change after a single processing operation. Within each route, the target, seed schedule, resolution, cameras, and evaluation remain fixed.

For paired differences, the report gives the mean, median, standard deviation, win rate, and bootstrap 95% confidence interval. With only four pairs, these intervals describe the observed candidate set and should not be read as population-level proof across arbitrary themes or weapons.

4.3 Seam-correction and selection-policy sub-ablations

The seam-correction table compares each formal Route B texture immediately before and after the locked correction. This isolates the effect of the seam operation without changing source content.

A separate edge-width sweep uses one unchanged earlier technical composition and tests widths 0, 2, 4, 8, 12, 16, and 24 pixels. The hard conditions are $E_{\text{asset}} \leq 0.01$ and multi-view retention of at least 95%. The same candidate pool is ranked by:

1. feasibility, Pareto dominance, and deterministic objective order; and
2. an equal 50/50 normalized weighted score.

This sweep supports claims about operating-point and policy behavior. It is not used to claim that the earlier artwork is visually superior to the final generated candidates.

4.4 Three-checkpoint live case

The interaction case begins with a dragon theme and the AK-47 adapter. The Agent produces four directions; the user selects *Treasured Relic*. Four landscape artwork variations are generated and mapped at low cost. One failed source attempt, containing large empty regions and isolated clusters, is rejected and regenerated locally rather than restarting the complete run. The user selects `artwork_04` after viewing the source with its left, right, and top previews. Formal 2048 processing then reuses the selected source exactly.

This case is not merged with the procedural formal A/B/C pool. Its purpose is to test the complete human-Agent interaction, real image-generation backend, checkpoint persistence, candidate-local retry, and final export contract.

4.5 Adapter transfer case

The M4A4 case reuses the same theme assets and core algorithms with zero additional image-generation calls. Only the adapter, official mesh hash, canonical axes, UV topology, and seam calibration change. The M4A4 mesh contains 45,422 triangles, 39,881 UV coordinates, 1,747 UV islands, and 11,341 seam pairs. The test asks whether the pipeline executes and where the AK-47 thresholds cease to transfer.

4.6 Evaluation criteria

The report uses the following primary measures:

- asset-seam error E_{asset} (lower is better);
- multi-view score S_{view} (higher is better);
- multi-view retention relative to the uncorrected candidate;
- locked Agent score for Route C accept/rollback decisions;
- changed-pixel fraction and changed pixels outside the target halo;
- successful completion of the three blocking human checkpoints;
- final texture resolution, colour mode, export identity, and file hash.

Automatic readability is reported only for explanation. It is not a substitute for a human preference study, and no claim is made that it measures general aesthetic quality.

Table 2: Formal paired A/B results. Lower asset-seam error is better; higher texture and multi-view scores are better.

Route	Candidate	Asset seam	Multi-view	Texture	Selected
A	01	0.286821	0.752696	0.761178	No
A	02	0.252571	0.792474	0.779769	Yes
A	03	0.243767	0.767358	0.770474	No
A	04	0.296737	0.763398	0.776778	No
B	01	0.000473	0.819614	0.765960	No
B	02	0.000635	0.818653	0.749159	No
B	03	0.000617	0.833842	0.755199	No
B	04	0.000796	0.844312	0.751902	Yes

Table 3: Paired Route B minus Route A summary ($n = 4$).

Metric	Mean	Median	Std.	Win rate	Bootstrap 95% CI
Asset-seam improvement	0.269344	0.269142	0.022281	100%	[0.247543, 0.291145]
Multi-view improvement	0.060124	0.066701	0.020440	100%	[0.036363, 0.077307]

4.7 Implementation and reproducibility

The implementation uses Python 3.13, NumPy, Pillow, OpenCV, SciPy, PyTorch, Transformers, Diffusers, and Streamlit. The local diffusion acceptance used an NVIDIA RTX 4060 Laptop GPU with 8 GB VRAM. The repository contains a deterministic procedural backend, a pinned SD-Turbo backend, provider adapters, the Agent runtime, command-line scripts, and a thin Streamlit client. At the report freeze, 100 unit tests, Python byte-code compilation, configuration parsing, and headless Streamlit startup had passed.

The public repository is `https://github.com/DID-FIELD/SDSC6002-SkinSmith`. Valve geometry, model weights, and API keys are intentionally excluded. Setup documentation explains how authorized users obtain external assets and configure local credentials.

5 Results

5.1 RQ1: weapon-space composition

Table 2 reports the four paired candidates. Route B reduces asset-seam error from 0.2438–0.2967 to below 0.0008 for every pair. It also improves the multi-view score for every pair, despite a modest reduction in the texture-only score. Candidate 04 is selected within Route B.

The paired summary in table 3 shows a mean asset-seam improvement of 0.26934 and a mean multi-view improvement of 0.06012. The win rate is 100% on both measures. The bootstrap intervals remain above zero for the four observed pairs.

The formal seam-correction operation reduces the mean Route B seam error by 99.62%. Before correction, the four values range from 0.14986 to 0.17639; after correction, all are below 0.0008 and pass the 0.01 threshold. This establishes that the low final seam error is not merely inherited from the source artwork.

(a) Route A candidate 04

(b) Route B candidate 04

Figure 3: Mapped left, right, and top views for the same candidate index in the formal experiment. Route A behaves as an all-over pattern; Route B reconstructs a continuous composition after UV fragmentation.

These results answer RQ1 positively for the tested AK-47 candidates. The conclusion is bounded to this asset, theme family, renderer, and four-pair sample.

5.2 RQ2: failures exposed by mapped evaluation

The live dragon case provides direct examples of why source images are insufficient. One attempted artwork passed technical image checks but was rejected because its subjects formed isolated clusters around large dark regions. Such a composition is fragile under cropping and UV fragmentation. After a local retry, all four candidate sources were mapped before selection.

For the selected source, automatic review reports source fulfillment 0.9386, left readability 0.6016, right readability 0.6016, and top readability 0.3993. Only eight of sixteen planned elements exceed the visibility threshold. The top view is the weakest because of its narrow projected area, while the source score is much higher. This difference is the central observation for RQ2: mapping reveals loss of readable elements, view imbalance, component-specific cropping, and seam behavior that cannot be inferred reliably from the rectangular artwork.

The route comparison in figure 5 illustrates a second failure mode. The generated Route A image has a high multi-view score of 0.8952, but its asset-seam error is 0.1822 and it fails the deployment constraint. A strong aggregate visual score is therefore not evidence of a technically valid texture.

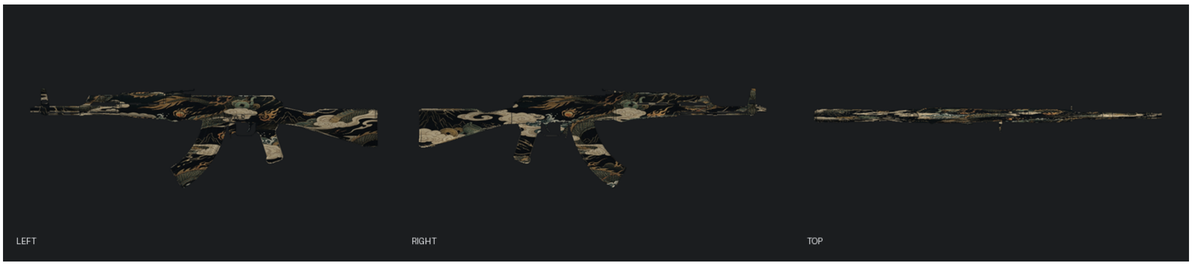
5.3 RQ3: constrained selection and rollback

Figure 6 shows the expected trade-off in the independent edge-width study. Increasing the safety width reduces seam error but gradually removes multi-view detail. Widths 8, 12, and 16 are feasible and Pareto-optimal. Width 24 has a lower seam error but falls below 95% retention.

The policy ablation is decisive. Equal weighted ranking selects width 4 because it retains a multi-view score of 0.70324, but its seam error is 0.01660 and therefore invalid. Constraint-first/Pareto selection chooses width 8, with seam error 0.00552 and multi-view score 0.69697. The weighted approach hides a hard violation behind a small visual gain.

All four formal Route C cases execute two local corrections. The selected attempts change 2.33–2.93% of atlas pixels and change zero pixels outside the target component plus halo. Their best locked-score changes are -0.00110, -0.00110, -0.00082, and -0.00088, all below the required +0.01. Consequently, every final C result equals B. In the live dragon case, the best correction changes the score by -0.02196 and is also rolled back.

(a) Human-selected landscape source artwork



(b) Formal Route B left, right, and top reconstruction

Figure 4: The selected source and its mapped result are related but not interchangeable evidence. The narrow top view and UV fragmentation remove or compress details that appear clear in the source.

RQ3 is therefore supported in two ways: feasibility-first selection rejects a candidate preferred by a soft score, and the acceptance gate prevents a lower quality correction from replacing its baseline. The result does not show that refinement improves quality; it shows that attempted refinement is bounded.

5.4 RQ4: three-checkpoint collaboration

The live Agent completes the intended interaction:

1. the theme is expanded into dragons, clouds, waves, treasure, flaming pearls, mountains, lightning, mist, jade, and architectural fragments;
2. four directions are presented, and *Treasured Relic* is selected;
3. four artworks are generated and mapped, and `artwork_04` is selected from its source-plus-three-view card.

The run uses six image calls, one local retry, and one refinement round. It ends in the

(a) Route A: strong visible activity, invalid asset seam

(b) Route B: valid seam and continuous mapped composition

Figure 5: The final live case separates visual activity from deployment feasibility.

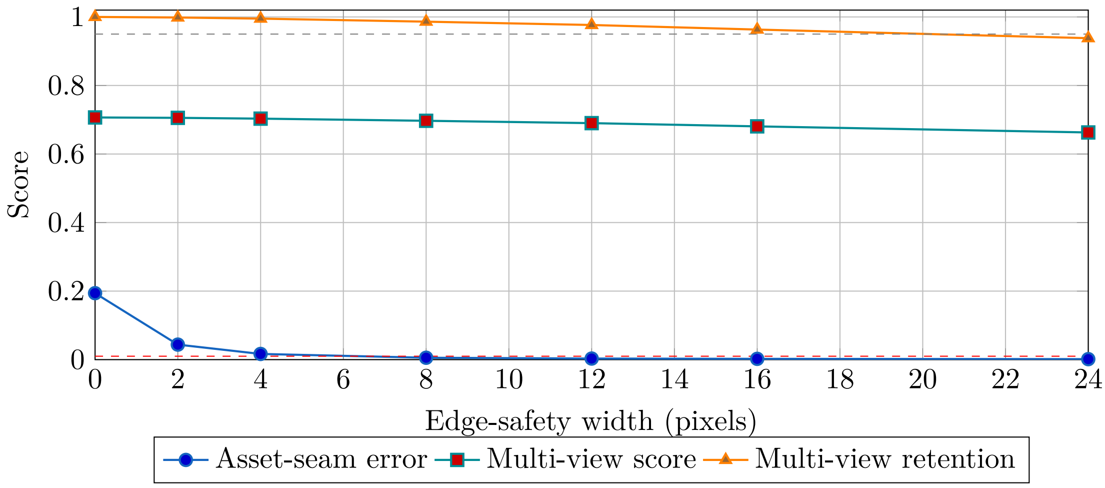


Figure 6: Independent edge-width sweep. Width 8 is the selected feasible operating point. The two dashed lines show the seam and retention thresholds.

`completed` state and exports a 2048 by 2048 RGB PNG and TGA with identical decoded pixels. The final Route B result has asset-seam error 0.000778, multi-view score 0.816620, and total score 0.829123. The source hash and final TGA hash are preserved.

This case answers RQ4 at the level of functional workflow: a user can begin from a compact theme, make three distinct decisions, and receive a formally processed texture without authoring a production prompt. It does not constitute a usability study; no completion-time or user-satisfaction comparison was conducted.

5.5 RQ5: adapter-based portability

The M4A4 adapter executes geometry mapping, continuous composition, UV baking, seam measurement, rendering, and evaluation with no new image-generation call. Figure 7 confirms that the same conceptual pipeline reconstructs a mapped texture on different geometry.

Transfer is only partial. The best joint candidate reaches seam error 0.00994 but retains 94.02% of the raw multi-view score, below the AK-47 threshold of 95%. The M4A4 contains 1,747 UV islands and 11,341 seam pairs, compared with 928 islands and 4,999 pairs for the AK-47. More aggressive seam treatment improves continuity but removes too much visible detail.

RQ5 is therefore answered by a boundary rather than a universal success: planning, composition, bake, rendering, metrics, and logging transfer; mesh, UV, axes, camera, material, and correction calibration remain adapter-specific.

Figure 7: M4A4 adapter transfer. The core method is reused, but the denser UV topology requires different seam calibration.

6 Discussion

6.1 Interpretation of the findings

The strongest result is not that Route B produces a more attractive square texture. Its flat atlas is often less legible than Route A because it stores pieces of a continuous composition. The relevant result is that the composition reconstructs on the weapon while satisfying a true asset-seam constraint. This changes the unit of evaluation from the generated rectangle to the mapped asset.

The paired experiment supports that representation choice for the tested AK-47 candidates. All four Route B candidates improve both seam and multi-view performance. The seam gain is expected in part because Route B explicitly contains an asset-edge operation, but the simultaneous multi-view gain is important: continuity was not purchased by simply flattening all visible detail. The separate before/after table further isolates the seam correction and shows that the operation, rather than the generator alone, is responsible for the low final error.

The live case complicates a simple "higher score is better" interpretation. Route A obtains a higher multi-view score than the selected Route B texture, yet fails the asset-seam threshold by a large margin. This is precisely the situation in which feasibility must precede preference. The edge-width ablation reaches the same conclusion under a fixed candidate pool: a small gain in multi-view score causes the weighted policy to choose an invalid candidate.

The rollback results are also informative. None of the formal Route C attempts improves the locked score, and the live correction is distinctly worse. Reporting these outcomes avoids presenting the Agent as an automatically self-improving system. The contribution is narrower and more defensible: the Agent can identify a target, produce local alternatives, prove locality, and decline to replace a better baseline.

6.2 Metric appraisal

The evaluation stack deliberately contains several measures because no single metric captures deployable visual quality. Asset-seam error has a clear technical interpretation, but a very low value can be achieved by over-smoothing boundaries. Multi-view retention protects against that failure, although it still summarizes contrast, saturation, detail, coverage, and consistency rather than aesthetic quality.

The component-detail balance metric failed on high-contrast weapon-space candidates: several Route B values collapsed to zero and made the locked Agent score lower even when seam and multi-view performance improved. For this reason, the component-balance and aggregate-score differences are not used as evidence that Route A is superior. They are retained as a documented metric limitation and as motivation for reporting primary measures separately.

Mapped-element readability has a similar boundary. The selected artwork scores highly at the source and substantially lower on the top view, which is useful diagnostic information. However, it depends on model-based semantic judgments and view weighting chosen for this case study. The human artwork decision remains authoritative.

6.3 Internal and external validity

Several controls strengthen internal validity: fixed assets and cameras, shared seed schedules within the formal comparison, exact source reuse after human selection, machine-readable logs, isolated sub-ablations, and deterministic rendering. The renderer, UV bake, seam metric, selection policy, and locality gate also have regression tests.

The main threat is sample size. Four formal pairs are sufficient to expose consistent behavior in the preserved candidate set but not to estimate performance over the space of themes, generators, or weapons. Route-specific generation logic also means the comparison evaluates two complete design strategies rather than a single-factor image transformation. This is appropriate for the research question, but the result should not be interpreted as a causal estimate of one isolated algorithmic component.

External validity is limited by one complete weapon, one principal creative case, one software renderer, and a small set of backends. The M4A4 result is useful because it fails the joint threshold: it shows that adapter portability is not the same as parameter portability. Broader claims would require several weapon families, game material systems, viewing conditions, and users.

6.4 Human-Agent interaction limitations

The three checkpoints solve a structural problem: they separate interpretation, creative direction, and mapped artwork. They also make disagreement recoverable. A user can edit a theme before generation, reject an unwanted direction, or reject an artwork after mapping without discarding the complete run.

The present study verifies this contract technically, but does not compare it with a one-prompt baseline through a user study. Future work should measure task time, number of revisions, perceived control, preference confidence, and the frequency with which source-only and mapped choices disagree. Such a study would provide stronger evidence for RQ4 beyond successful execution.

6.5 Project management and methodological evolution

The registered topic, *Latent Vision → Image Generation*, was intentionally broad and emphasized image-to-image workflows, diffusion models, conditioning, and creative visual outputs. Supervisor guidance later superseded that original scope: given the remaining delivery window, completing the technical development of an image-processing Agent was considered sufficient, and the Agent did not need to preserve the original Latent Vision framing. The team therefore treated SkinSmith as a focused implementation of the approved image-processing Agent direction rather than as an attempt to cover every objective in the original topic description.

Work then progressed through dated milestones: deterministic generation and seam repair; real OBJ rendering; local diffusion; UV and seam diagnostics; weapon-space composition; formal A/B/C experiments; provider-backed generation; the reusable Agent runtime; the three-checkpoint client; and academic production.

Several changes were driven by failed evidence rather than by the original plan. Direct border blending was removed after it produced a visible mirrored frame. An early mask-aware route was retained as negative evidence because it changed component appearance without creating a coherent whole-weapon composition. Per-component dragon generation was abandoned after it produced semantically wrong source parts and framed panels. A legacy UV assumption was rejected after deployment mismatch. These corrections delayed the interface, but reduced the risk of polishing a workflow whose asset contract was wrong.

The group used five connected work packages with one integration lead. YUAN Ye served as project lead and integration owner, defining the system architecture and coordinating the shared Agent contract. CHEN Yuhong owned creative planning and literature synthesis; Ben Tangjie owned generation-backend integration and reproducibility; Wang Zhengye owned asset, UV, and rendering validation; and WANG Lihui owned evaluation, experiment reporting, and submission quality control. Each work package contributed method decisions, implementation or configuration, experimental verification, and documentation to the final system. This structure gave every member an assessable technical contribution while retaining one owner for cross-module consistency.

6.6 Responsible use, provenance, and rights

The project uses generative models as experimental components. Structured theme and style planning in the accepted live case used `gemini-3.1-flash-lite`; source artwork used `gemini-3.1-flash-image`; and SD-Turbo provided a pinned local baseline. Prompts, model identifiers, validation traces, and source hashes are recorded. Credentials are supplied through local environment variables and removed from evidence.

The system is restricted to original themes and explicitly prohibits copied skins, logos, brands, teams, protected characters, text, watermarks, and imitation of a named living artist. Generated sources are reviewed before mapping. Valve geometry is used only as an external case-study asset and is not redistributed in the repository. The report does not claim ownership of Valve assets, affiliation with Valve, or readiness for Steam Workshop publication.

These safeguards do not resolve every concern. The training data of external image models are not controlled by this project, and automated filters can miss similarity or embedded content. Any use beyond the academic prototype would require model-licence review, independent rights checking, stronger moderation, game-specific terms review, and a clear appeals process for rejected or disputed content.

6.7 Practical implications

Three practical lessons follow from the study:

1. A source image should not be approved before it is mapped. Candidate cards should bind the source to the views on which it will be judged.
2. A deployment constraint should not be averaged into a general quality score. Feasibility and preference answer different questions.
3. Agent refinement should have a budget and a rollback rule. In creative pipelines, a plausible correction can still remove valuable detail.

These lessons extend beyond weapon skins to other assets whose appearance is stored in fragmented atlases, including props, clothing, vehicle parts, and product visualizations. The geometry and deployment adapters would change, but the evidence and decision structure would remain relevant.

7 Conclusion

This project investigated how a compact visual theme can be converted into a mapped, technically valid, and reproducible game-asset texture. The resulting SkinSmith prototype combines three human checkpoints with asset-aware generation, deterministic UV baking, multi-view evaluation, hard feasibility rules, and one bounded refinement round.

For RQ1, the four-pair AK-47 experiment shows that composing in continuous weapon space and baking into the existing atlas improves both true asset-seam and multi-view performance relative to the tileable baseline in every observed pair. For RQ2, mapped evaluation exposes view-specific loss, internal seam failure, and the difference between an attractive source and a usable texture. For RQ3, the selection-policy ablation and the Route C results show that hard constraints and rollback prevent acceptance of infeasible or lower-scoring candidates. For RQ4, the live run demonstrates a complete path from a short theme through theme, direction, and mapped-artwork decisions to a 2048 PNG/TGA export. For RQ5, the M4A4 case shows that the core pipeline transfers through an adapter, while UV topology and correction parameters remain asset-specific.

The main conclusion is methodological: image generation should be treated as one stage in an asset pipeline, not as the endpoint. A credible generative Agent must show what happens after mapping, preserve the user's authority where metrics are weak, distinguish feasibility from preference, and retain the evidence behind acceptance and rollback.

Future work should evaluate more weapon and non-weapon assets, add game-engine render capture, improve semantic and perceptual measures, learn adapter-specific seam correction, and conduct a user study of the three checkpoints. Those extensions would test whether the observed benefits generalize beyond the present case without weakening the bounded claims established here.

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A Reproducibility

A.1 Repository

The project repository is:

https://github.com/DID-FIELD/SDSC6002-SkinSmith

External Valve assets, model weights, and API credentials are not included. The repository setup guide describes how to obtain authorized geometry and configure local dependencies.

A.2 Core commands

```
.\.venv\Scripts\python.exe -m unittest discover -s tests -v
.\.venv\Scripts\python.exe -m compileall -q src scripts streamlit_app.py
    tests
.\.venv\Scripts\python.exe scripts\smoke_test.py
.\.venv\Scripts\python.exe scripts\real_model_smoke_test.py
.\.venv\Scripts\python.exe scripts\abc_ablation_formal.py
.\.venv\Scripts\python.exe scripts\build_report_ablation_tables.py
.\.venv\Scripts\python.exe -m streamlit run streamlit_app.py
```

The exact arguments for provider-backed runs and checkpoint resume are documented in the repository because they depend on local credentials and preserved run state.

B Evidence index

Table 4: Machine-readable evidence used in the report. Paths are relative to the repository root.

Claim or artifact	Preserved evidence
Formal A/B/C candidates	runs/abc_ablation_true_weapon_space/ablation_log.json
Paired statistics	runs/report_ready_ablation_tables/table_abc_paired_summary.csv
Formal seam correction	runs/report_ready_ablation_tables/table_formal_b_seam_correction.csv
Edge-width sweep	runs/report_ready_ablation_tables/table_edge_safety_sweep.csv
Selection-policy comparison	runs/report_ready_ablation_tables/table_selection_policy_comparison.csv
Formal Route C attempts	runs/report_ready_ablation_tables/table_route_c_attempts.csv
Three-checkpoint live case	runs/agent_dragon_multicandidate_v1/agent_run_result.json
Artwork alternatives	runs/agent_dragon_multicandidate_v1/artwork_candidates.json
Mapped readability	runs/agent_dragon_multicandidate_v1/execution/mapped_element_readability.json
M4A4 transfer	runs/m4a4_game_asset_adapter_transfer/transfer_final.json

C Formal Route C attempts

Table 5: Best correction attempt for each formal Route B candidate. All modifications remain inside the target component and transition halo.

Candidate	Changed fraction	Outside-halo pixels	Score change	Accepted
01	0.028670	0	-0.001097	No
02	0.029309	0	-0.001095	No
03	0.023314	0	-0.000816	No
04	0.027806	0	-0.000880	No

D Milestone summary

Table 6: Development milestones and corrective decisions.

Stage	Completed capability	Evidence-based adjustment
Foundation	Procedural candidates, periodic repair, metrics, logs	Mirrored border blending was replaced after visual failure.
Real asset loop	OBJ/UV rendering and fixed three-view evaluation	Flat texture quality was no longer treated as sufficient evidence.
UV diagnosis	Welded seam graph, masks, asset binding	Incompatible legacy UV assumptions were rejected.
Weapon-space route	Canonical frame, geometry maps, continuous composition	Mask-only remapping was retained as negative evidence and replaced.
Formal evaluation	A/B/C, seam and policy sub-ablations, M4A4 transfer	Metrics with invalid behavior were reported with caveats.
Reusable Agent	State, budgets, retries, checkpoints, exact source reuse	Per-component generation was replaced by a master-artwork workflow.
Client and delivery	Thin Streamlit client, replay, report evidence	Feature development was frozen before academic production.

E Project scope record

The project was registered as *Latent Vision → Image Generation*. The original topic description covered broad image-to-image workflows and several possible creative outputs. The supervisor subsequently advised narrowing the project to a completed technical image-processing Agent because of the remaining schedule, and confirmed that the work did not need to remain the original Latent Vision project. The SkinSmith title and scope record the final approved research object; the registered topic remains on the cover for administrative traceability.

F Group contribution statement

Table 7: Detailed group work-package ownership. YUAN Ye served as project lead and integration owner; all members held a substantive technical and academic responsibility.

Member	Work-package ownership	Technical and academic contribution
YUAN Ye	Project leadership, system architecture, and integration	Defined the scoped research problem and Agent architecture; implemented and integrated the runtime, checkpoints, asset-aware execution, weapon-space method, selection and rollback logic, CLI, and Streamlit client; coordinated formal runs, cross-module debugging, evidence freeze, and final manuscript integration.
CHEN Yuhong	Creative planning, prompt methodology, and literature	Developed and reviewed the ThemePack/StylePack planning logic, art-direction differentiation, source-content constraints, and human-checkpoint rationale; synthesized literature on controllable generation, human-AI creativity, and responsible generation; evaluated source-art compliance and qualitative mapped failure cases.
Ben Tangjie	Generation backends, experiment reproducibility, and runtime	Integrated and verified procedural, SD-Turbo, and API generation workflows; maintained model and prompt configuration, retry and checkpoint records, environment and execution instructions, provider trace redaction, and reproducibility checks for candidate generation and replay demonstrations.
Wang Zhengye	Game-asset adapters, UV processing, and multi-view rendering	Developed and validated asset binding, mesh/UV diagnostics, camera orientation, seam correspondence, left-right-top rendering, texture export inspection, and the M4A4 transfer case; reviewed mapped candidate cards and asset-specific portability limits.
WANG Lihui	Evaluation methodology, statistical analysis, and deliverables	Developed and verified texture, asset-seam, multi-view, feasibility, and refinement evaluation; prepared formal A/B/C and sub-ablation tables, interpreted negative results and limitations, maintained claim-to-evidence checks, and coordinated report, poster, and presentation quality assurance.